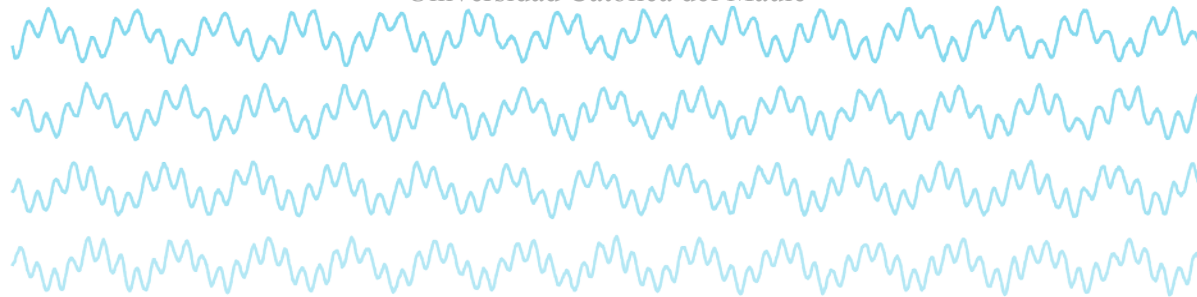


EEG for Neuroscience Research (I)

From Neurons to Scalp Signals

Dr. Boris Lucero
CINPSI Neurocog
Universidad Católica del Maule





Facultad de Ciencias de la Salud

Faculty of Health Sciences

The Neuropsychology and Cognitive Neurosciences Research Center



CINPSI
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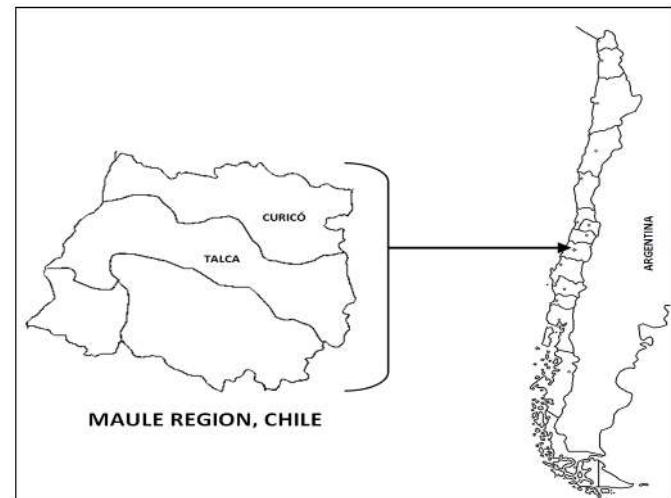
About CINPSI Neurocog

Led by Boris Lucero, PhD, CINPSI Neurocog is based in the Technological Sciences Building at the Catholic University of Maule.

Our research advances neuropsychology and cognitive neuroscience through the integration of behavioral approaches and neurophysiological measures, including electroencephalography (EEG) and transcranial magnetic stimulation (TMS), with a particular focus on identifying biomarkers of neurophysiology, brain function and cognition.







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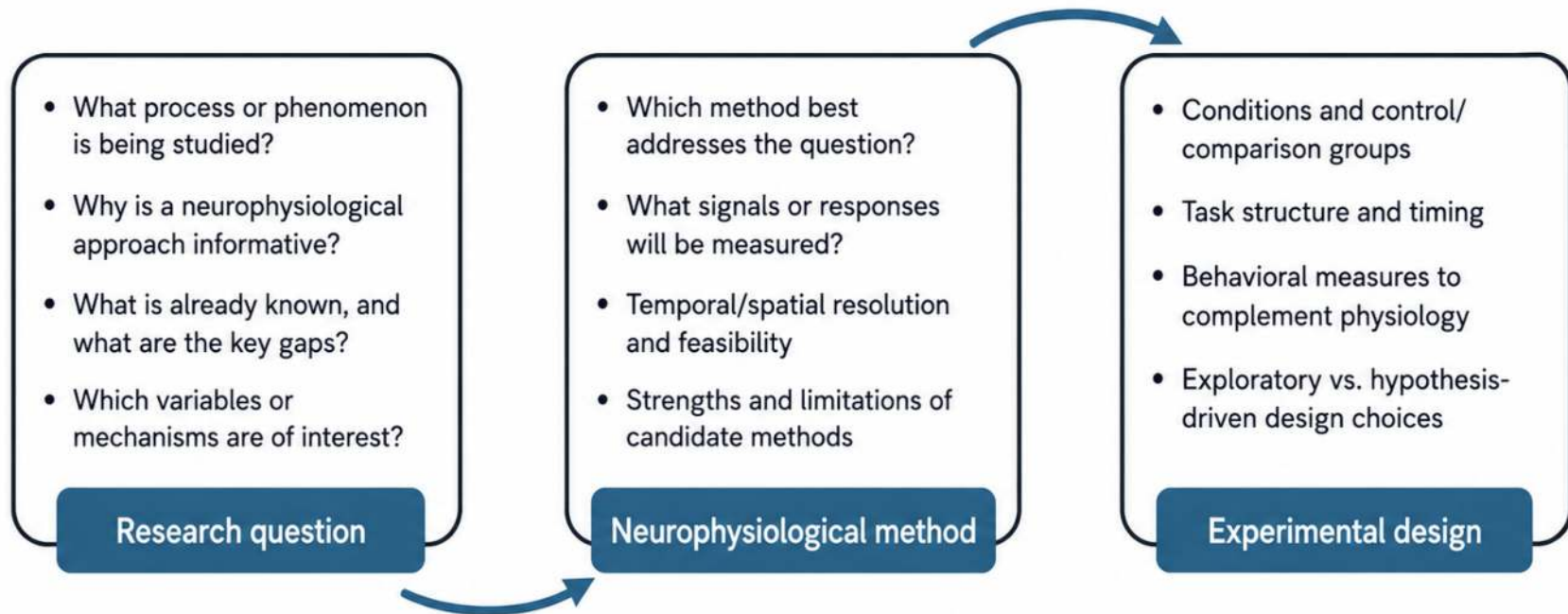
www.cinpsineurocog.com

OUTLINE

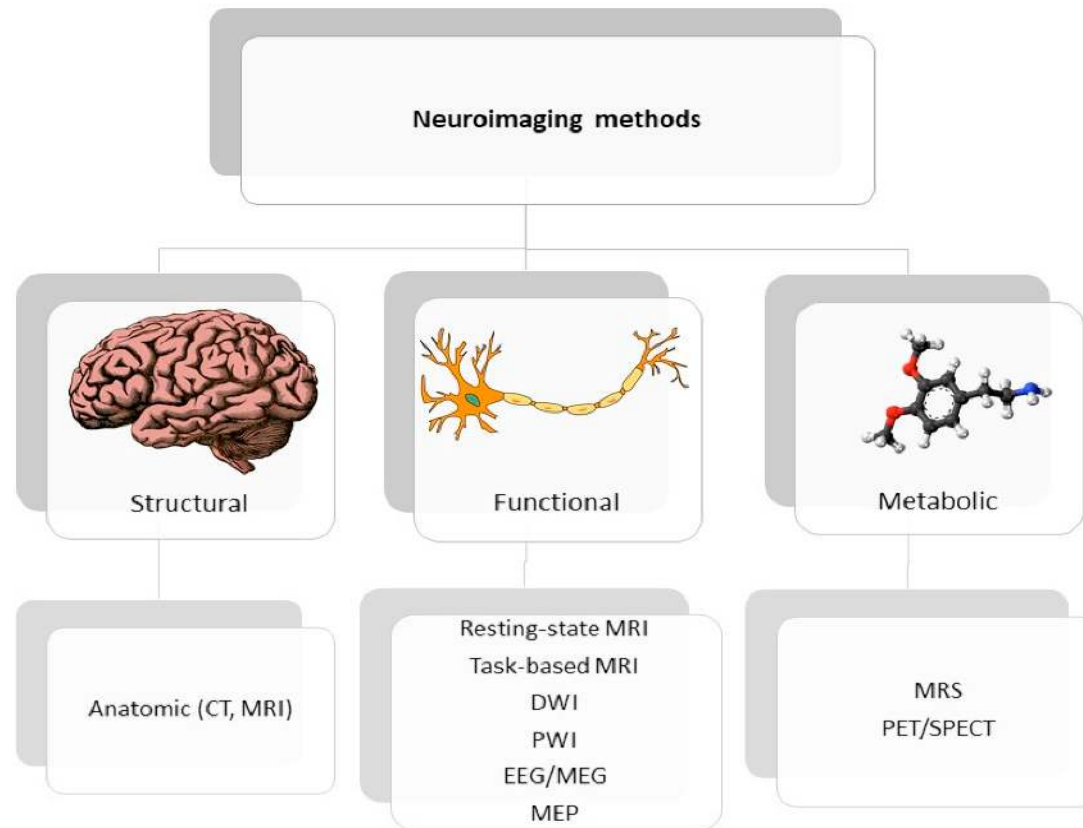
1. Why EEG in neuroscience research?
2. What does EEG measure?
Signal origin
3. EEG and oscillations
4. Signal acquisition
5. Signal quality



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Neurociencias Cognitivas



Adapted from: Ninaus, M., Kober, S. E., Friedrich, E. V., Dunwell, I., De Freitas, S., Arnab, S., ... & Neuper, C. (2014). Neurophysiological methods for monitoring brain activity in serious games and virtual environments: a review. *International Journal of Technology Enhanced Learning*, 6(1), 78-103.

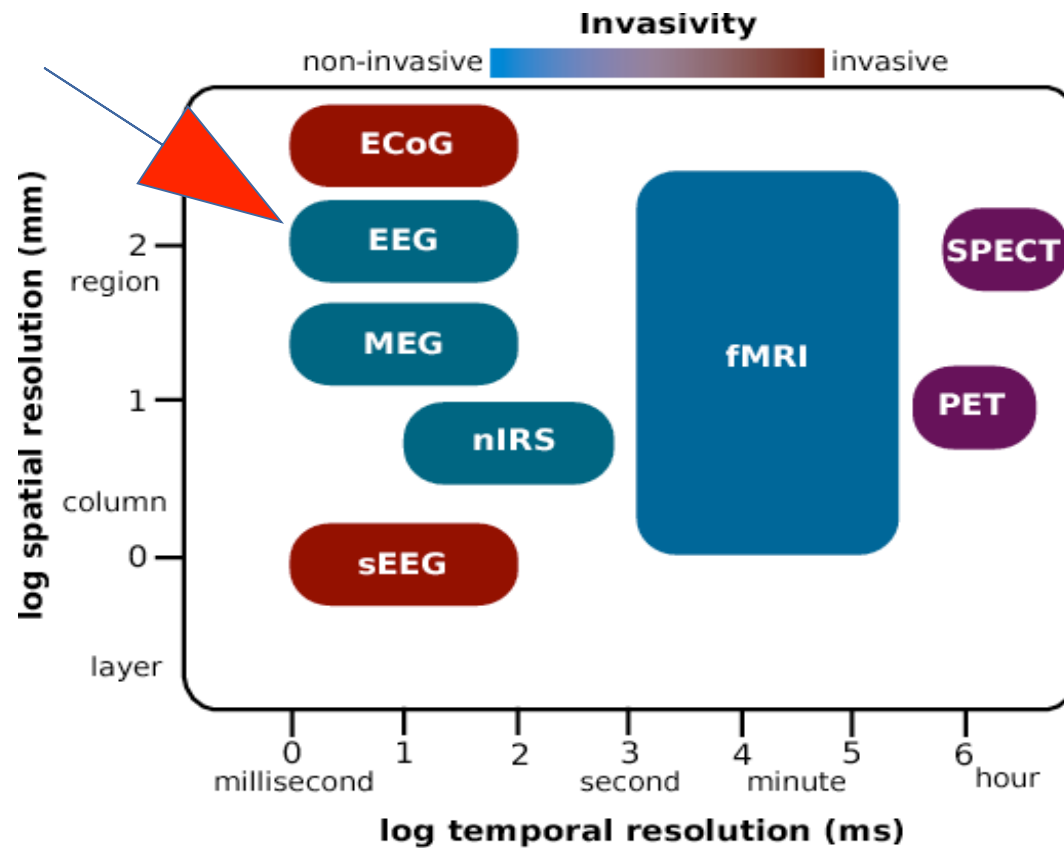


Włodarczyk, L., Cichon, N., Saluk-Bijak, J., Bijak, M., Majos, A., & Miller, E. (2022). Neuroimaging techniques as potential tools for assessment of angiogenesis and neuroplasticity processes after stroke and their clinical implications for rehabilitation and stroke recovery prognosis. *Journal of Clinical Medicine*, 11(9), 2473.

Why EEG?

Why EEG?





Włodarczyk, L., Cichon, N., Saluk-Bijak, J., Bijak, M., Majos, A., & Miller, E. (2022). Neuroimaging techniques as potential tools for assessment of angiogenesis and neuroplasticity processes after stroke and their clinical implications for rehabilitation and stroke recovery prognosis. *Journal of Clinical Medicine*, 11(9), 2473.

Why EEG? A direct, millisecond-scale window on population activity



Direct neural measure

Reflects electrical activity itself, not a hemodynamic proxy



Millisecond resolution

Tracks neural dynamics at the speed of cognition



Accessible

Low cost per session relative to MRI/MEG; scalable to large samples



Portable & tolerant

Usable in clinics, schools, workplaces, and field settings

Method	Signal	Temporal res.	Spatial res.
Scalp EEG	Summed currents	~1 ms	cm-scale
MEG	Magnetic fields of same currents	~1 ms	sub-cm–cm
fMRI (BOLD)	Hemodynamic proxy	~1–2 s	mm
LFP / spikes (invasive)	Local currents / action potentials	<1 ms	μm–mm

Trade-off to remember: EEG buys temporal precision and ecological reach at the cost of spatial certainty.

ELECTROENCEPHALOGRAPHY (EEG)

- EEG is a **non-invasive**, **direct** and **continuous** measure of the changes in electrical potentials **at the scalp** produced by neural activity in **REAL time**.

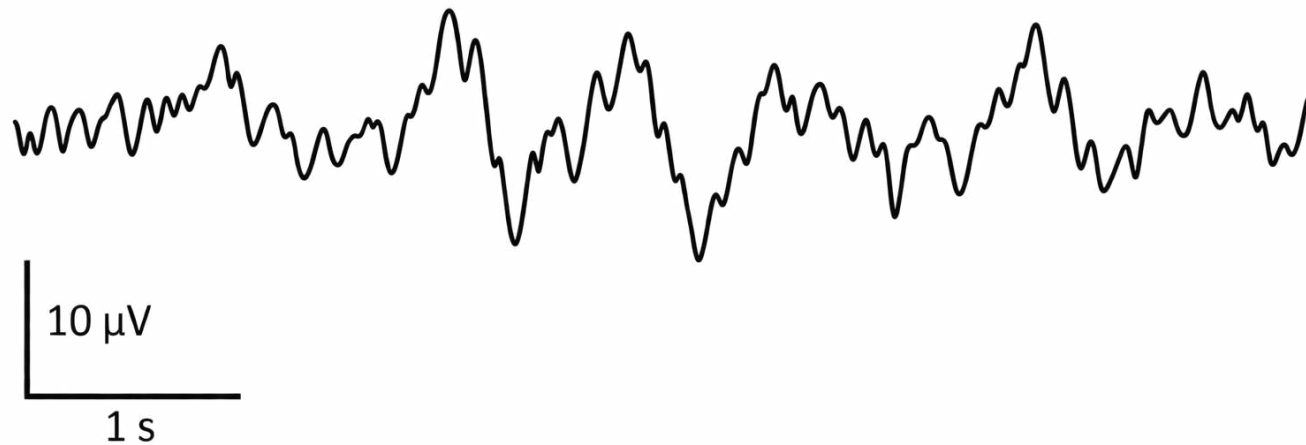


Brain Vision EEG
System



What does EEG measure?

Action potentials?



Videos:

<https://pursue.richmond.edu/products/animations/>

Understanding EEG: Understanding Potentials

This project was sponsored by PURSUE

Preparing Undergraduates for Research in STEM-related fields Using Electrophysiology

Cindy Bukach, University of Richmond, Catherine Reed Claremont McKenna College, Jane Couperus, Hampshire College

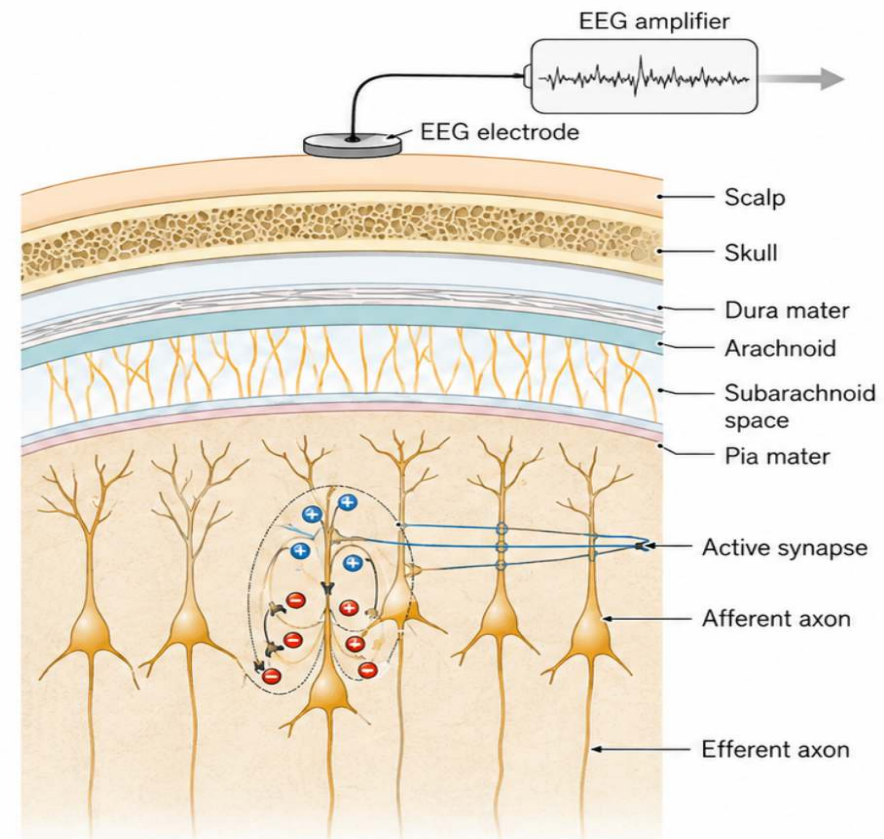
Supported by grants from NSF DUE 1625521, DUE 1625610, DUE 1626554



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Cindy Bukach, Jane Couperus, Catherine L. Reed

EEG

- Measures the brain's bioelectrical activity detected at the scalp surface through the use of electrodes.
- EEG signals arise from excitatory postsynaptic potentials (EPSPs) and inhibitory postsynaptic potentials (IPSPs) generated by neurons arranged in organized layers, or by dendrites whose apices are oriented perpendicular to the cortical surface (for example, pyramidal cells and cell columns in the visual cortex).



Understanding EEG: Formation of a Dipole

This project was sponsored by PURSUE

Preparing Undergraduates for Research in STEM-related fields Using Electrophysiology

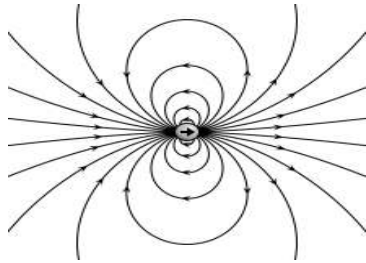
Cindy Bukach, University of Richmond, Catherine Reed Claremont McKenna College, Jane Couperus, Hampshire College

Supported by grants from NSF DUE 1625521, DUE 1625610, DUE 1626554

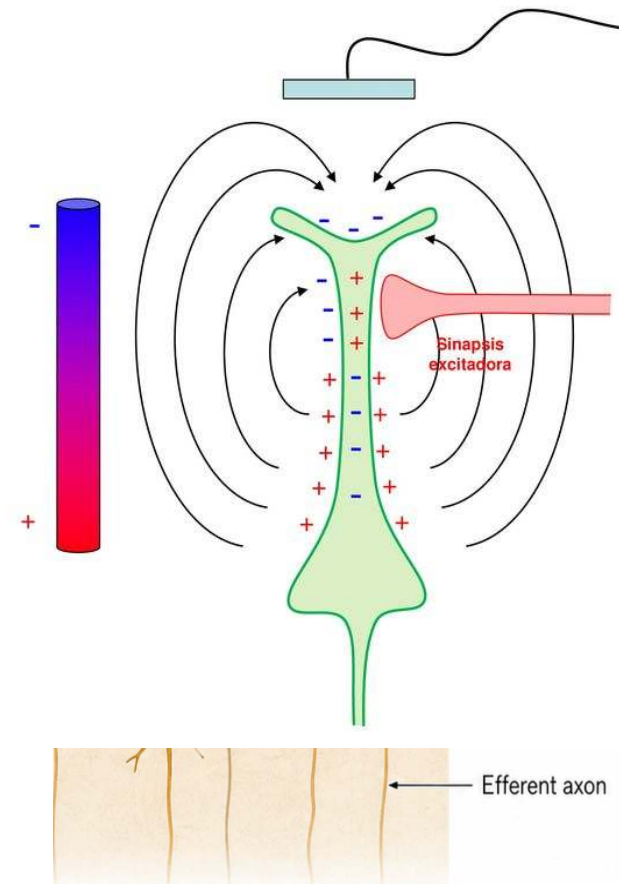
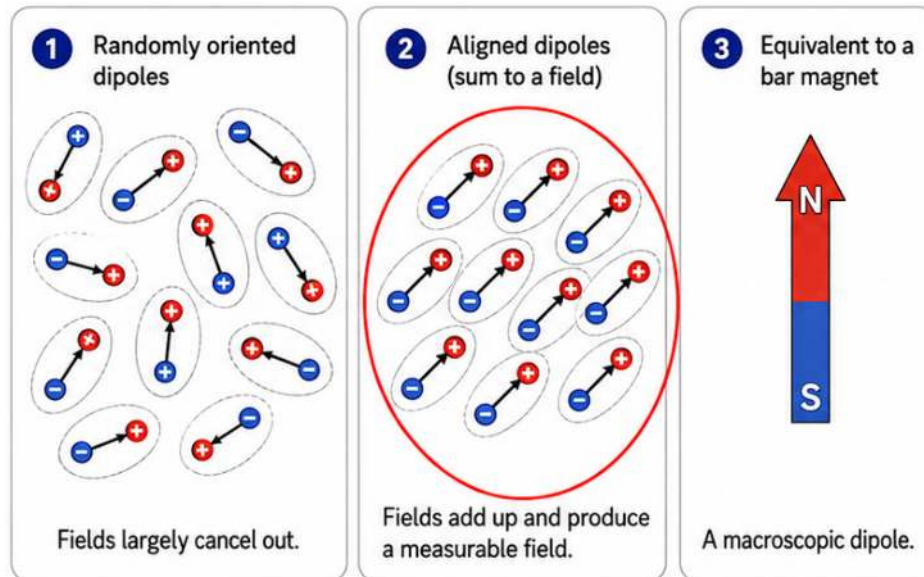


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Electric fields in the Brain



An **electric dipole** is a system of two nearby charges of opposite sign and equal magnitude that generates an electric field



Understanding EEG: Sensing Dipoles

This project was sponsored by PURSUE

Preparing Undergraduates for Research in STEM-related fields Using Electrophysiology

Cindy Bukach, University of Richmond, Catherine Reed Claremont McKenna College, Jane Couperus, Hampshire College

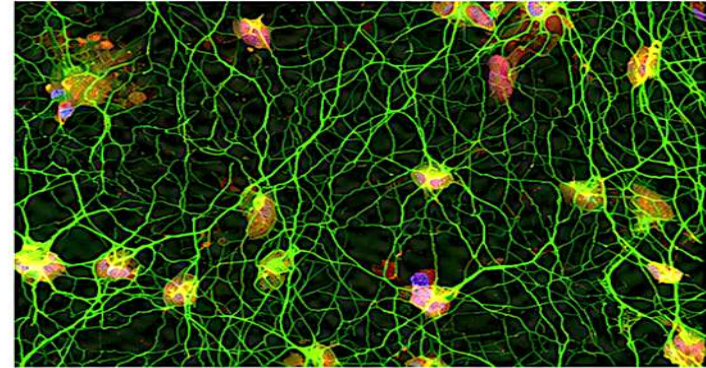
Supported by grants from NSF DUE 1625521, DUE 1625610, DUE 1626554



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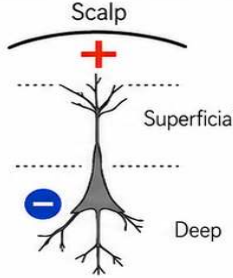
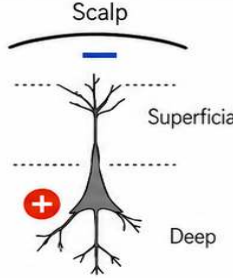
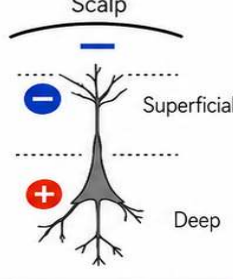
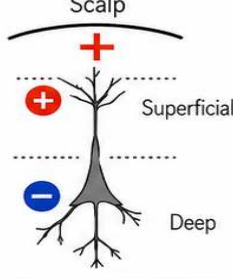
How strong is the electricity generated by a neuronal field?

- EEG is displayed as a continuous waveform with ongoing oscillations.
- The largest amplitudes can reach 150–200 microvolts.
- In other words:
- Millions of neurons acting together generate an electrical field.
- This electrical field is about 10,000 times smaller than that produced by an AA battery.



Scalp polarity depends on synapse location, not only on synapse type

Local intracortical polarity reflects sink/source at the synapse, but scalp polarity depends on whether the active pole is superficial or deep.

	EPSP	IPSP
Deep / middle cortical synapse	• Local effect: active sink at the synapse (negative extracellular potential). • Return current: superficial source. • Scalp polarity: positive. 	• Local effect: active source at the synapse (positive extracellular potential). • Return current: superficial sink. • Scalp polarity: negative. 
Superficial synapse	• Local effect: active sink at the synapse (negative extracellular potential). • Return current: deeper source. • Scalp polarity: negative. 	• Local effect: active source at the synapse (positive extracellular potential). • Return current: deeper sink. • Scalp polarity: positive. 

 Active sink (net inward current)
  Active source (net outward current)

Key point: Surface polarity alone cannot tell you whether the underlying synaptic activity is excitatory or inhibitory.

The scalp electrode samples a volume-conducted field; it does not mark the site where current exits the neuron.

From cortex to scalp: summation, volume conduction, attenuation

$\sim 10^4 - 10^7$

synchronously active neurons needed to produce a measurable scalp deflection

$\sim 6 - 10 \text{ cm}^2$

of coherently active cortex is a common order of magnitude for visible scalp EEG

$\times$ blur

skull and tissue conduct current diffusely, spatially smearing and attenuating the signal

- **Volume conduction: currents spread instantaneously through brain, CSF, skull, and scalp, every electrode sees a weighted mixture of many sources**
- Consequence 1: the forward problem is well-posed (sources $\rightarrow$ scalp map is computable), but the inverse problem is ill-posed (scalp map $\rightarrow$ sources has no unique solution)
- Consequence 2: an electrode label (e.g., Cz) names a scalp location, not a brain structure underneath it
- Consequence 3: signals mix across channels; this motivates source modeling and the connectivity caveats of Part II

Note: figures above are order-of-magnitude teaching values; exact numbers vary by source geometry and study.

Forward vs. inverse: why “where is it coming from?” is hard

FORWARD PROBLEM: well-

known sources + head model $\rightarrow$ unique scalp map

Given the generators and the conductive geometry of head tissues, the scalp distribution can be computed exactly.

INVERSE PROBLEM: ill-posed

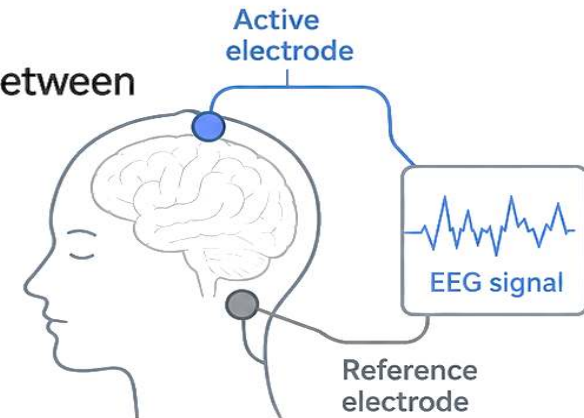
scalp map $\rightarrow$ infinitely many source configurations

Different generator sets can produce identical scalp maps, no measurement alone can decide between them.

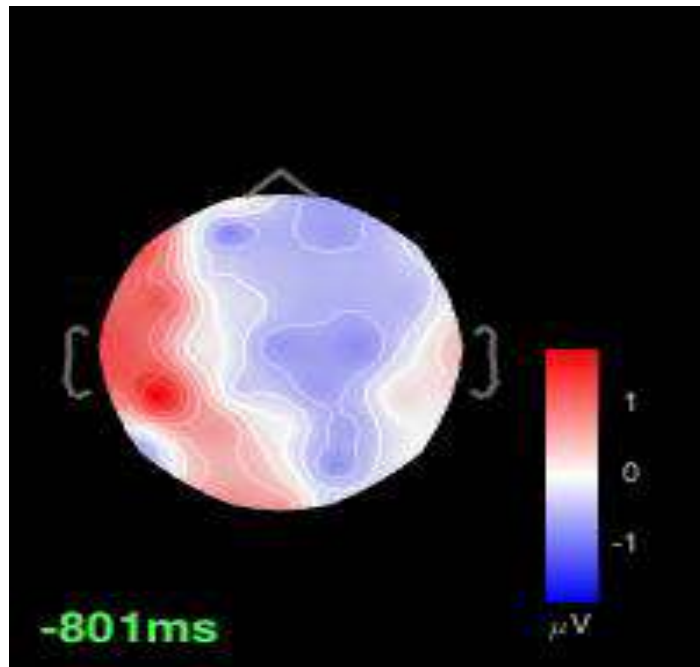
- **Source estimation therefore adds assumptions to choose among solutions: equivalent-dipole fitting (few focal sources), distributed models with smoothness or minimum-energy priors (LORETA-family, minimum-norm), and spatial-filter approaches (beamformers)**
- Quality depends on head model, electrode density and positions, and clean data.
- **Read source maps as model-based estimates with assumption-dependent uncertainty, never as photographs of brain activity**

In Summary

- EEG reflects postsynaptic potentials generated in the cortex by the synchronous activity of neuronal populations.
- A large number of neurons must be activated synchronously because the signal has to pass through several layers of non-neural tissue (meninges, intervening fluids, skull bone, and skin) before reaching the electrode.
- EEG detects the difference in electrical potential between an active electrode and a reference electrode.
 - placed on the scalp (*cephalic reference*)
 - placed at other body sites (mastoids, earlobes), considered electrically inert or, at least, lacking their own electrical activity (*non-cephalic reference*)

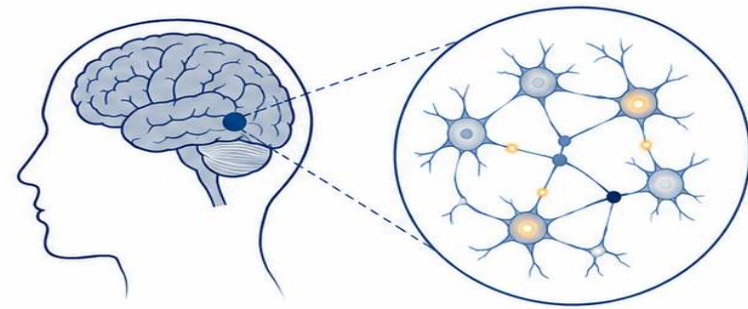
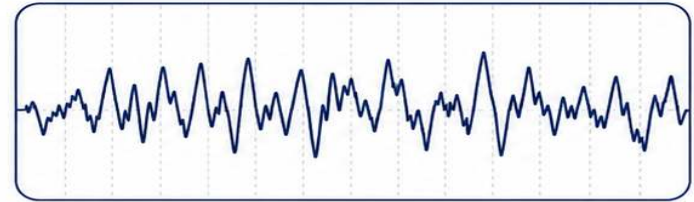


Brain: Oscillatory States



EEG and Oscillations

- EEG data contain rhythmic activity.
- This rhythmic activity reflects neuronal oscillations.
- Oscillations are fluctuations in the excitability of neuronal populations.
- Some oscillations are fast, others are slower, and some are very slow.
- Some oscillations last for long periods of time (seconds to tens of minutes), whereas others are more transient.
- Some oscillations change according to task events, whereas others appear not to be related to task events.

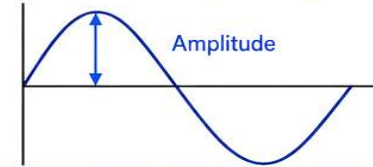


Components of Oscillations

- Oscillations contain three main pieces of information: frequency, power, and phase.

1. **Frequency:** The rate of oscillation, measured in hertz (Hz). It refers to the number of cycles per second and is the reciprocal of time. Therefore, 2 Hz means 2 cycles per second, 40 Hz means 40 cycles per second, and 0.1 Hz means 1 cycle every 10 seconds.
2. **Power:** The amount of energy in a frequency band. It corresponds to the square of the oscillation's amplitude.
3. **Phase:** The position along the sinusoidal wave at any given point in time, measured in radians or degrees.

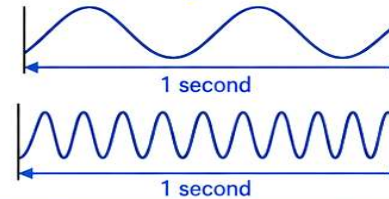
1. Amplitude (Power)



Amplitude is the maximum displacement from the baseline.

Power is proportional to the square of amplitude.

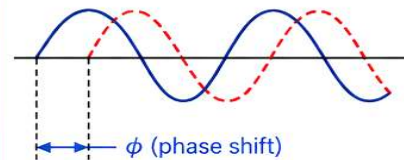
2. Frequency



Fewer cycles in 1 second
= lower frequency

More cycles in 1 second
= higher frequency

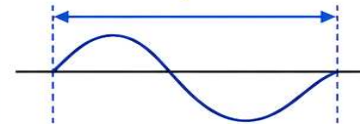
3. Phase



— Reference wave
- - - Phase-shifted wave

Phase (ϕ) is the horizontal shift between two waves, measured in radians (e.g., 0 to 2π) or degrees (0° to 360°).

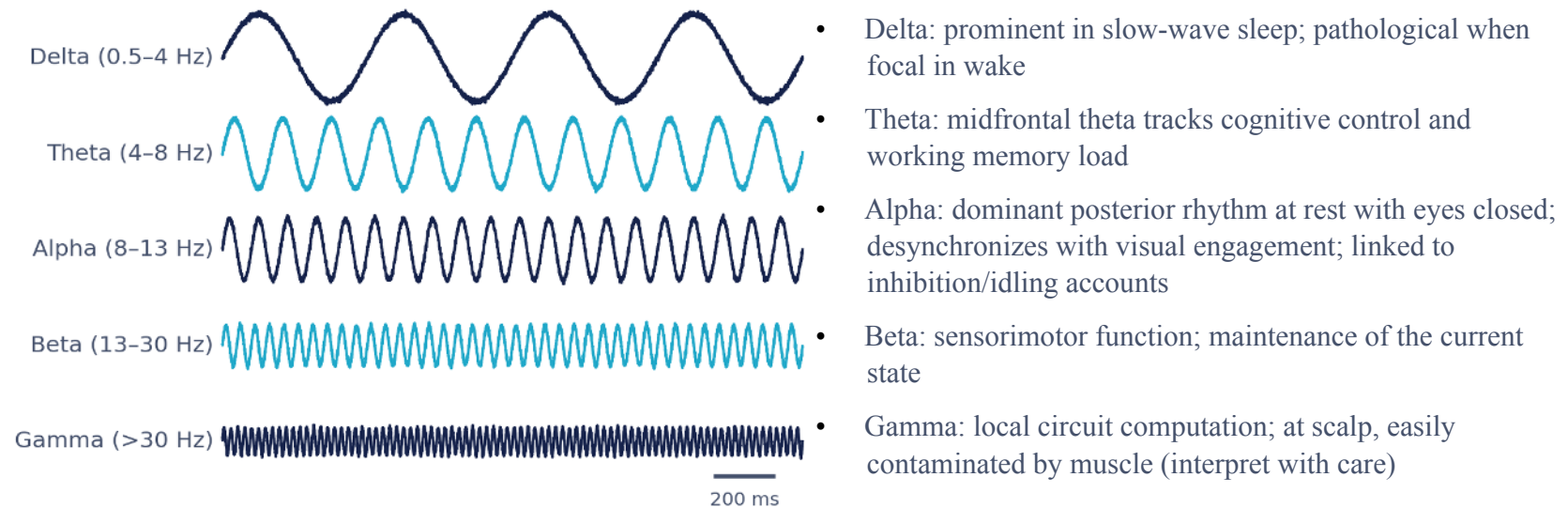
What is one cycle?



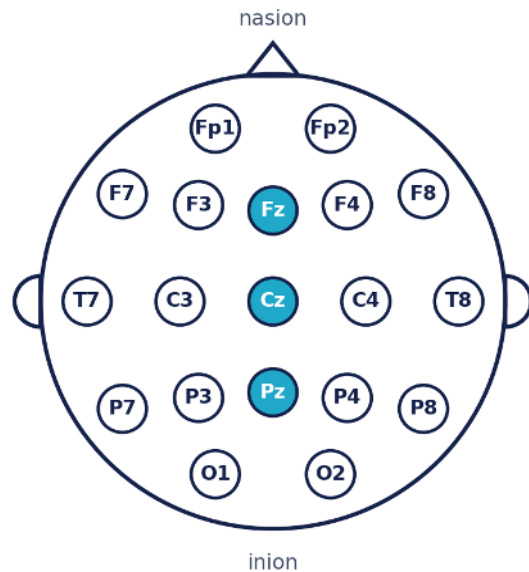
One cycle =
one complete oscillation.

From a point on the wave to the next identical point in phase (e.g., upward zero crossing to the next upward zero crossing).

Brain rhythms: the spectral vocabulary of EEG



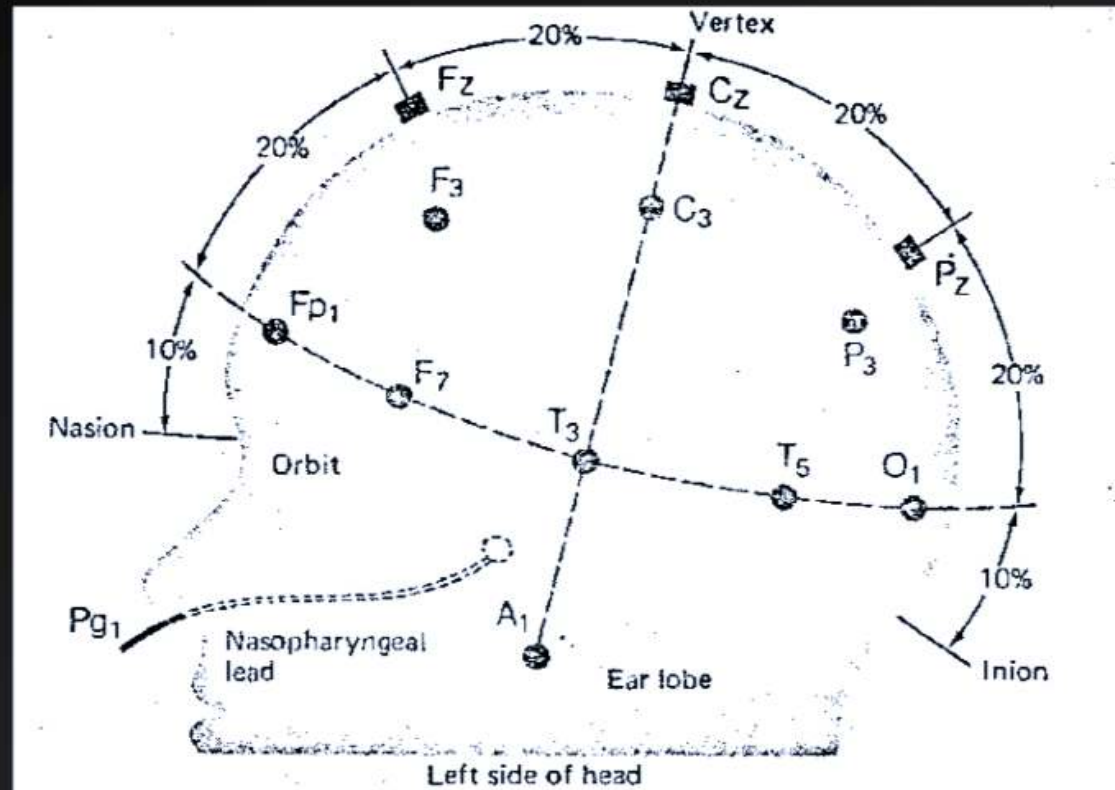
Recording EEG: electrodes, the 10–20 system, and the reference



International 10–20 layout (midline sites highlighted)

- **Standardized positions (10–20 / 10–10) make montages comparable across labs and studies**
- Wet Ag/AgCl electrodes remain the reference standard; gel or saline systems trade preparation time against signal stability; dry systems ease field use
- **Every channel is a potential DIFFERENCE: active site minus reference — there is no reference-free voltage**
- Common references: linked mastoids, average reference, Cz-online with offline re-referencing; the choice reshapes your waveforms and must be reported
- Ground electrode $\neq$ reference: it serves amplifier noise rejection
- Keep electrode–skin impedance low and, above all, balanced across channels

Recording EEG



BioSemi



1. Key features

- Electroencephalography (EEG) acquisition system
- Measures electrical potential
- Up to 264 channels (electrodes)
- Uses active electrodes
- High resistance to electrical noise

2. Active vs. passive electrode systems

Active electrodes	Passive electrodes
• A small preamplifier is built into the electrode • Less sensitive to cable noise and environmental interference • More tolerant of higher electrode impedance • Implication: easier setup and cleaner signals in many situations	• No local amplification at the electrode • Signal travels through the cable before amplification • More sensitive to noise and cable movement • Implication: usually requires lower impedance and more careful preparation

3. Handling precautions



- Never force the connectors or spill water or gel on them
- Avoid contact with metal objects
- Do not leave components submerged in water
- Do not use soap or detergent
- Do not connect non-approved electrical devices

Components of the BioSemi System

1. Core components

- A/D box: receives the signal directly from the electrodes and performs analog-to-digital conversion. This A/D box remains in the recording room with the participant.
- Active electrodes
- Pin-type electrodes:
 - These electrodes are connected to the participant's cap. They are flat ribbon cables that end in a connector used to connect to the A/D box.
 - Active electrodes amplify the signal at the head, which means good-quality signals can be obtained with higher impedance (typically <30 ohms, rather than <5 ohms for passive electrodes).



Pin-type electrode ribbon cable assembly

2. Flat electrodes

- These electrodes are used to record from flat body or facial surfaces:
- EOG (electro-oculogram) / EXG electrodes on the face
- EMG (electromyogram) electrodes on the body
- Reference electrodes

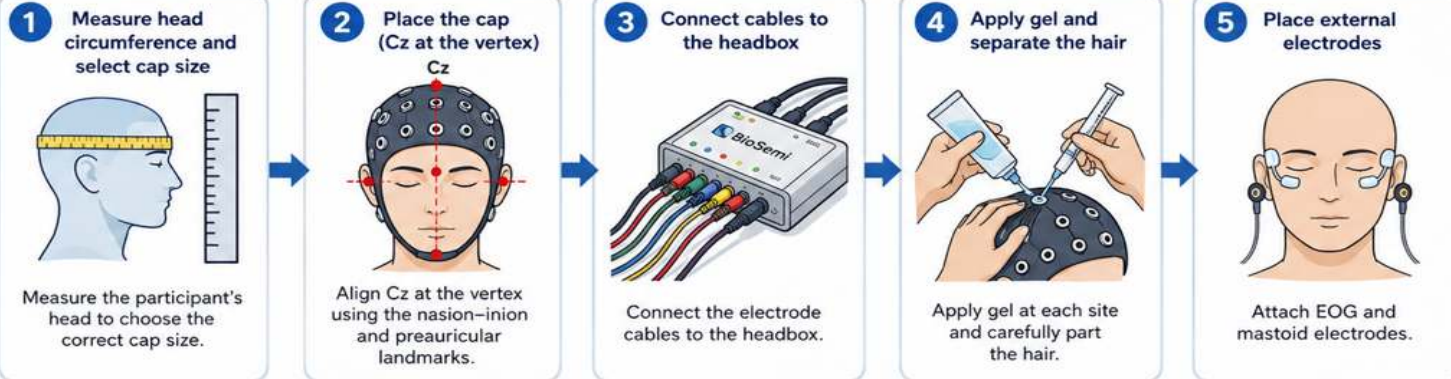
Materials

- Gel (SignaGel)
- Syringes (fine and curved tip)
- Alcohol, gel, or cleaning solution to clean the skin before placing electrodes on the face
- Measuring tape for head measurement
- Cap
- Chin strap to keep the cap in place
- Paper towels
- Adhesive rings (for placing EXG electrodes)
- Electrodes
- Shampoo, disposable towel, and brush

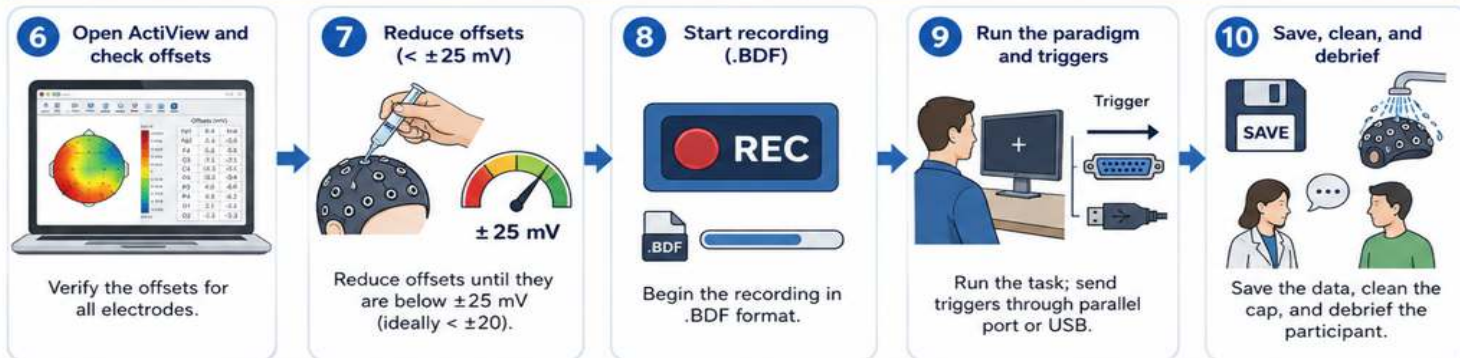


EEG Recording: Step-by-Step

Preparation



Recording



Signal and Noise: a Complicated Relationship

Signal-to-Noise Ratio (SNR)

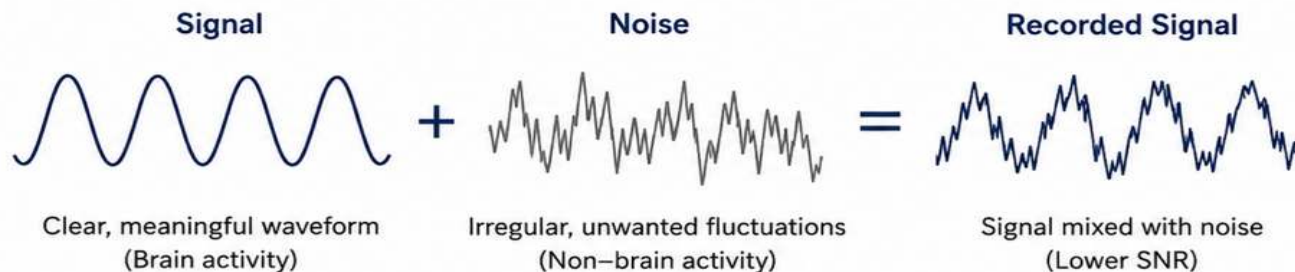
- A measure of how much signal is captured by the system relative to how much noise is present.

(Jackson & Boger, 2014)

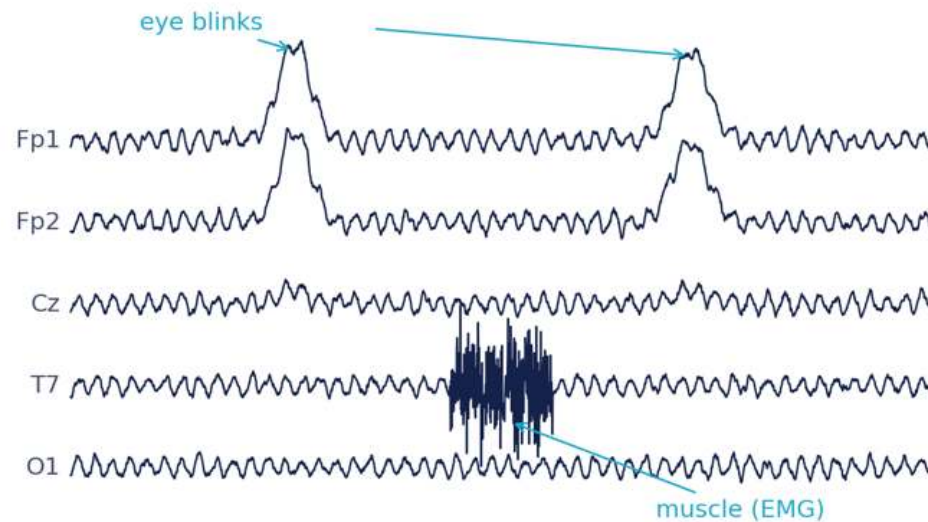
Noise

- “Refers to any voltage that is recorded and does not correspond to what the system is intended to measure; in other words, any voltage that does not represent brain activity.”

(Jackson & Boger, 2014)



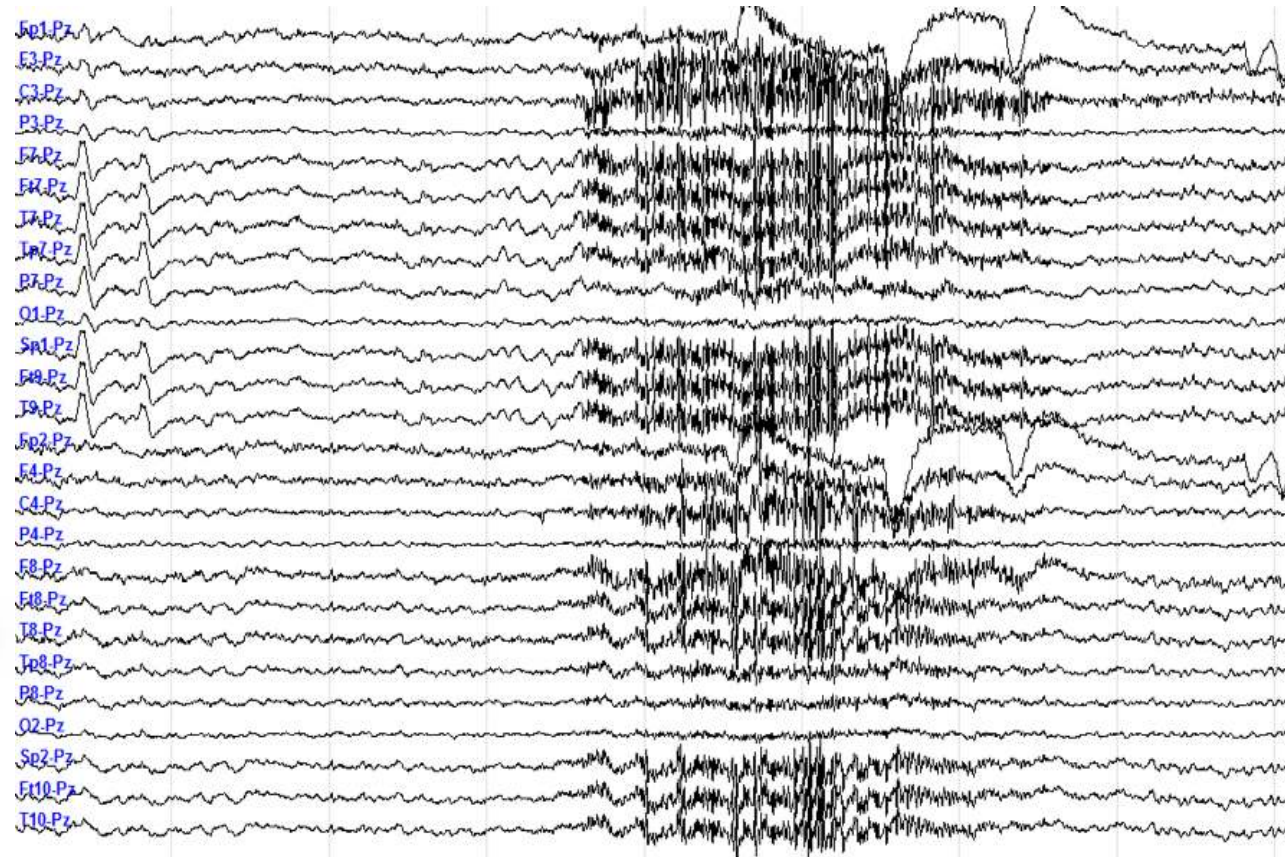
Artifacts: the biggest signals in your recording are not brain



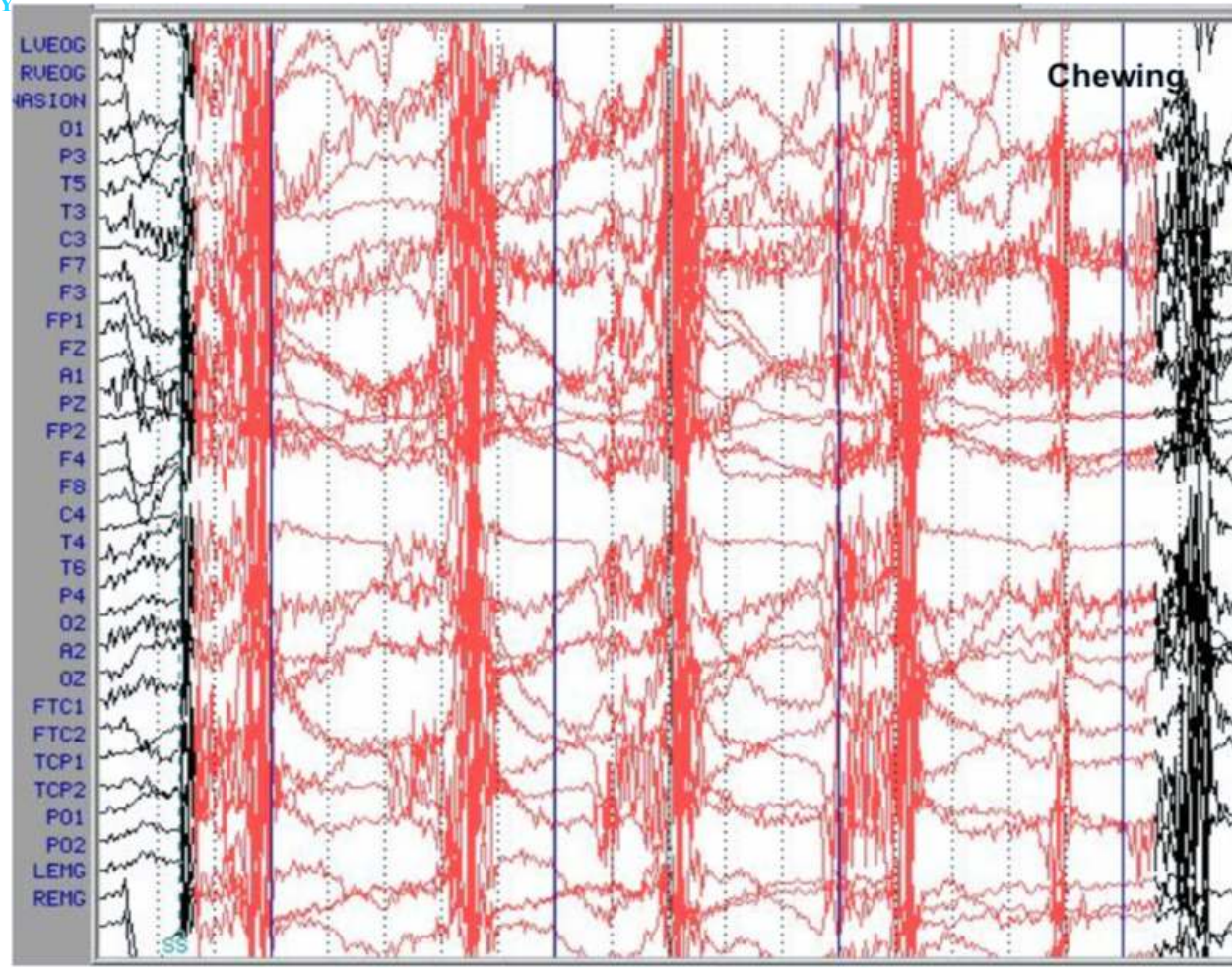
Simulated data for illustration

- **Ocular — blinks & saccades: large, frontal, stereotyped**
- Muscle (EMG) — broadband high-frequency; temporal/neck sites; the main threat to gamma claims
- Cardiac (ECG/pulse) — rhythmic, reference-dependent
- Line noise — 50/60 Hz and harmonics
- Electrode artifacts — pops, drift, poor contact, sweat
- **First defense is prevention at recording time; second is detection; only third is correction**

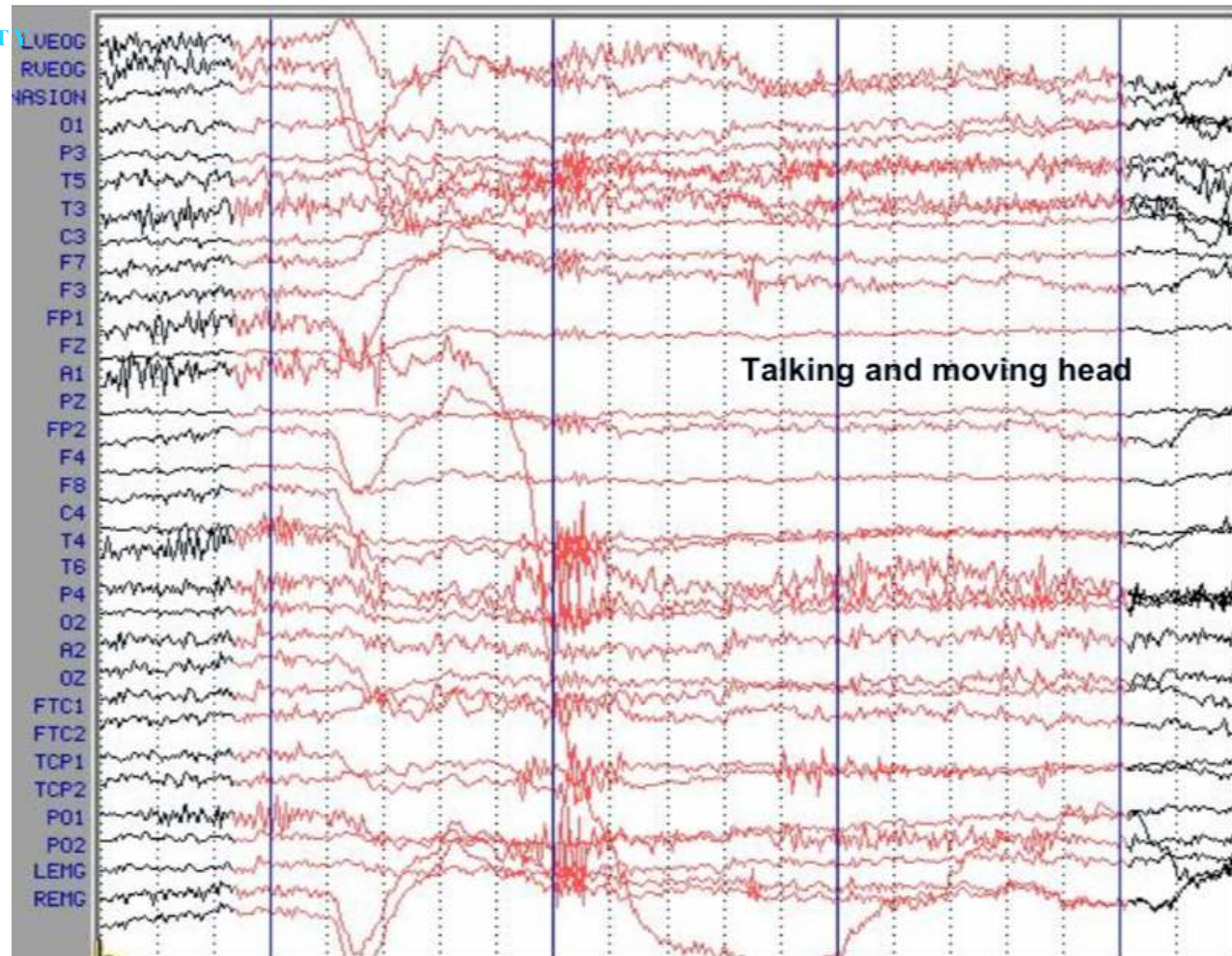
Muscle Noise



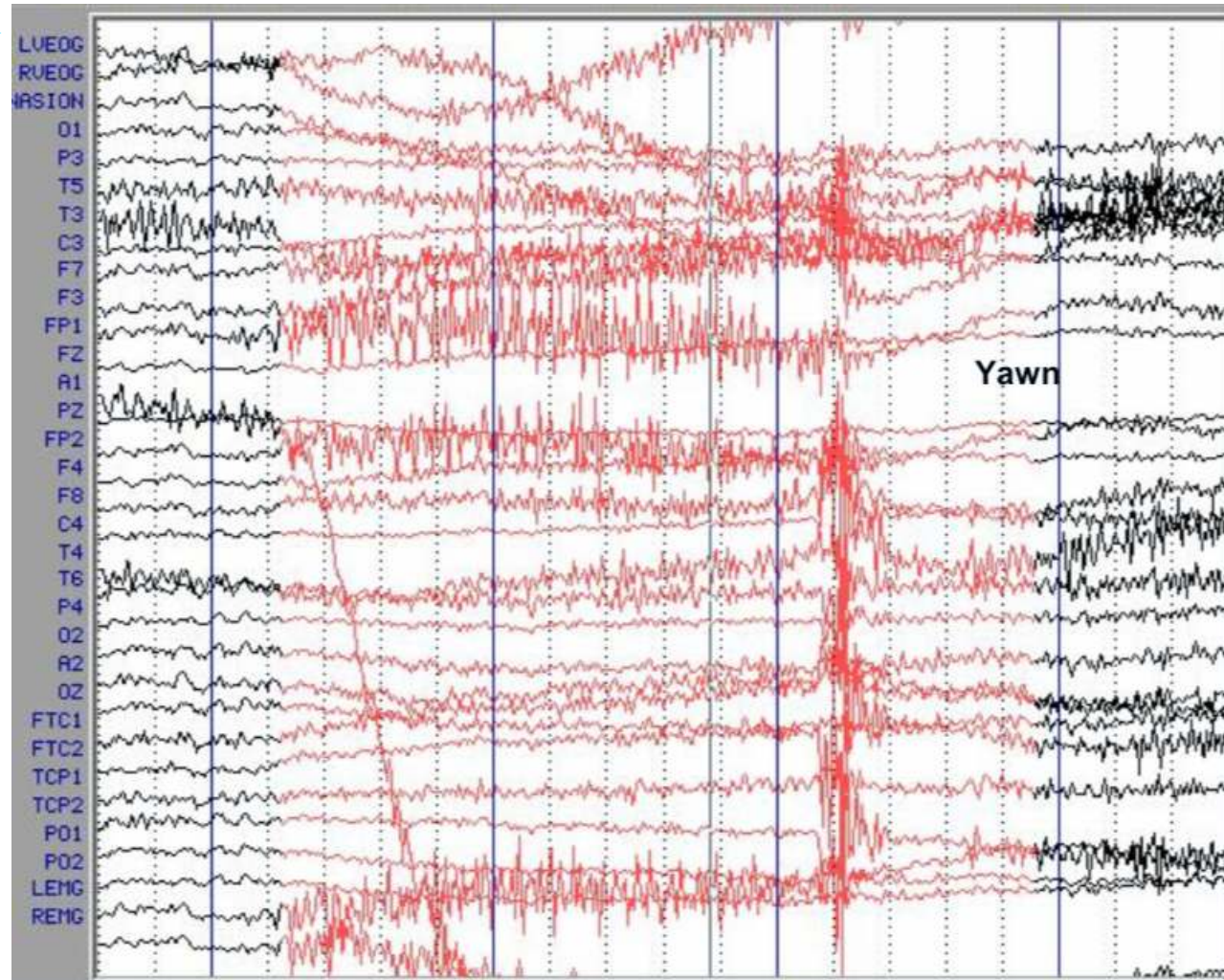
SIGNAL QUALITY



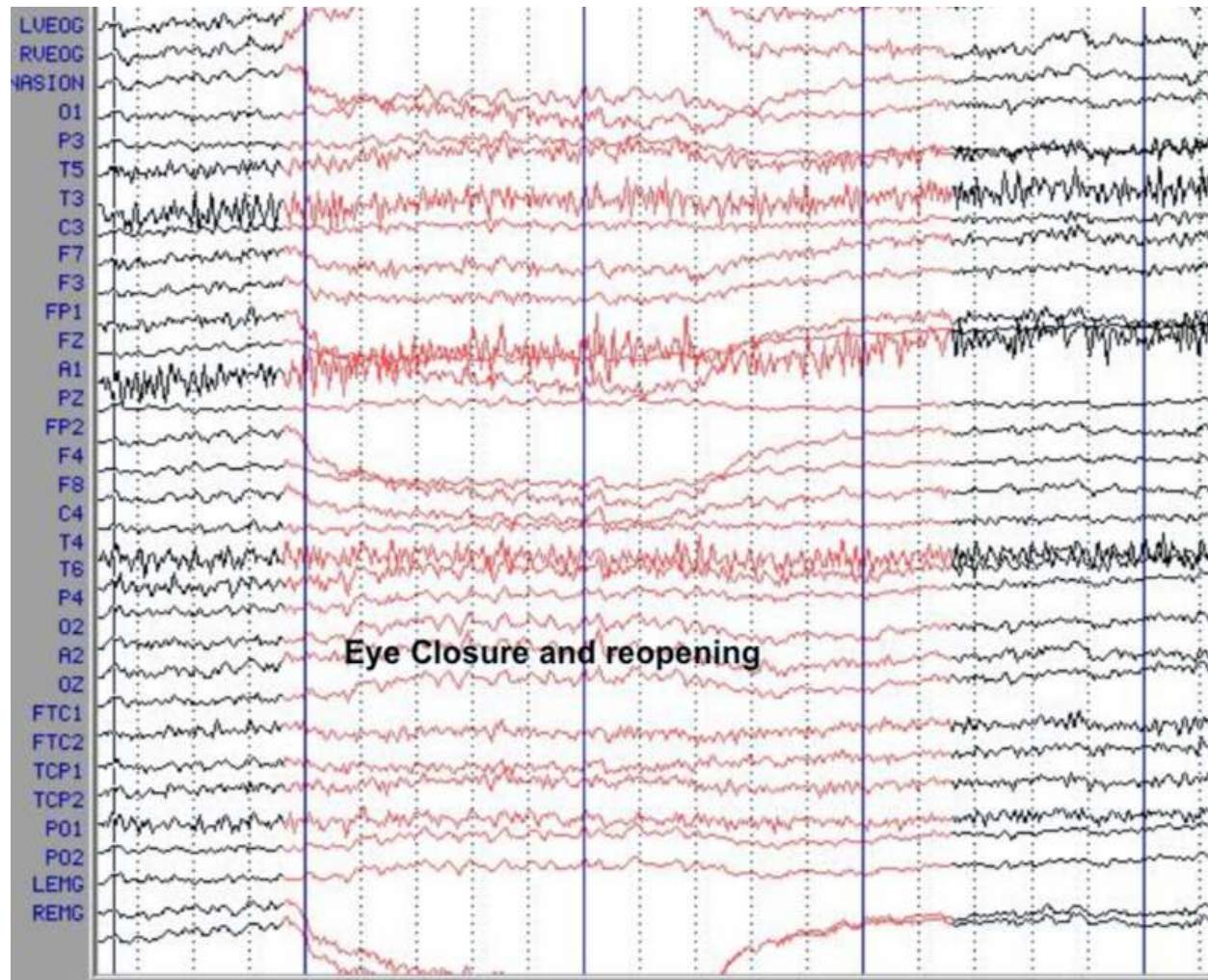
SIGNAL QUALITY



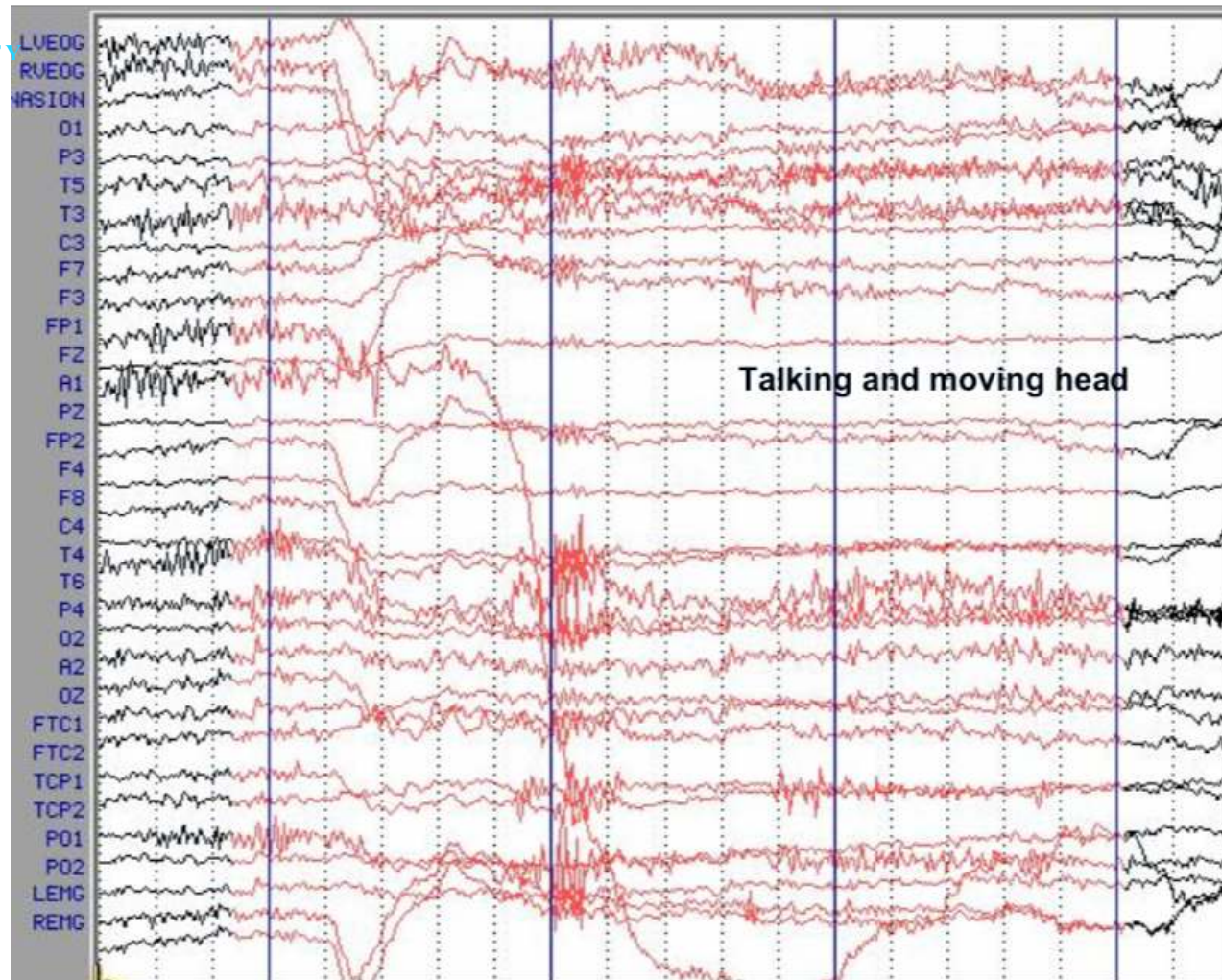
SIGNAL QUALITY



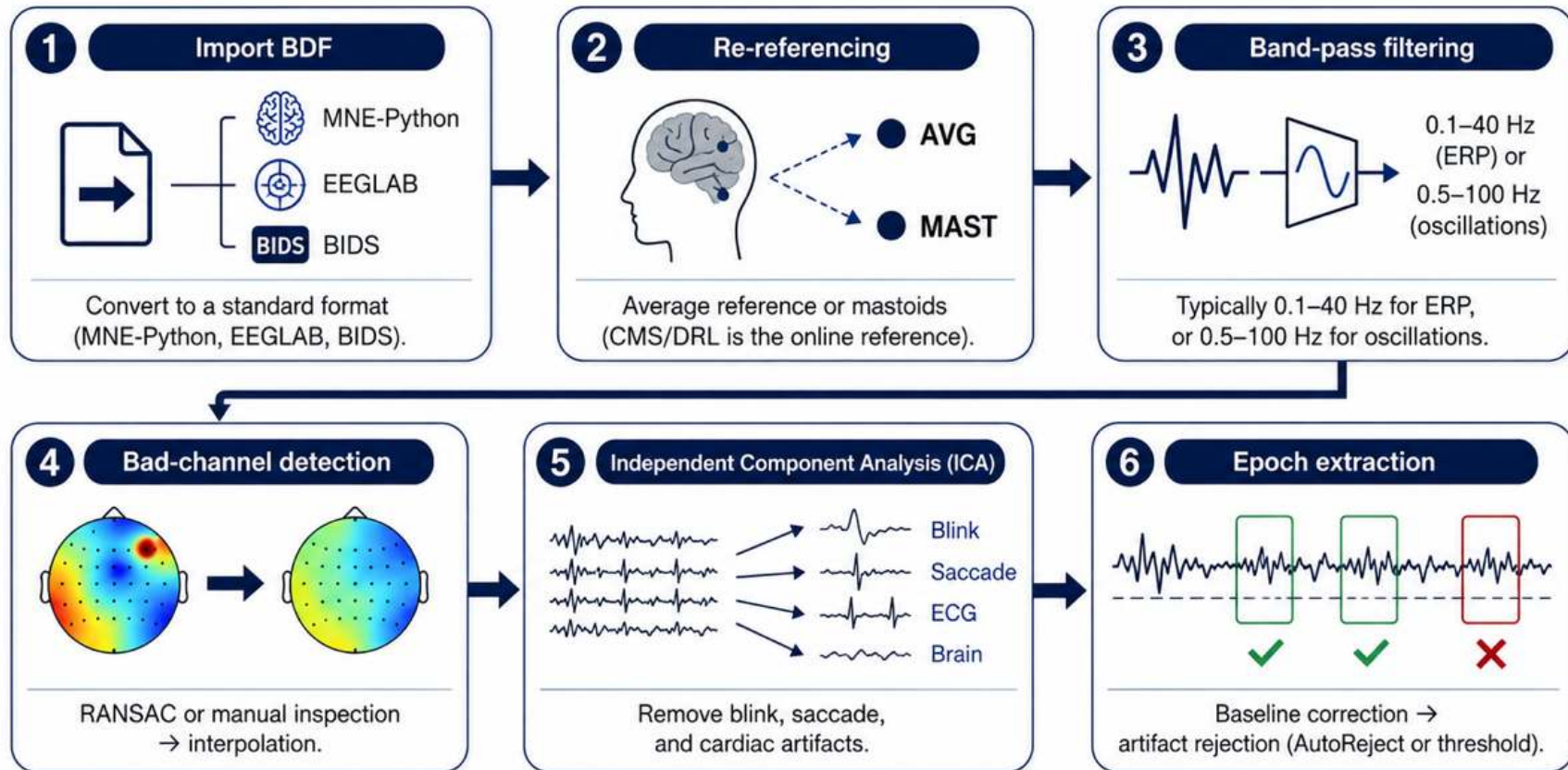
SIGNAL QUALITY



SIGNAL QUALITY



EEG Preprocessing Pipeline



Take-Home ideas

- EEG is the summed synaptic activity of synchronised, geometrically aligned pyramidal populations.
- Volume conduction mixes sources: scalp location is not brain source, and every voltage is relative to a reference.
- Acquisition and cleaning decisions set a hard ceiling on every later analysis.

Thank you

Next hour: Dr. Christ Devia

